

HELP DESK TICKET SLA & VOLUME FORECASTING

AI / ML CAPSTONE | Historical evidence. Reproducible research.

27,348

SCOPED TICKET RECORDS

4.67

TICKETS / DAY MAE | 30-DAY TEST

19.92%

SLA REFERENCE MET | HISTORICAL

1



FRAME THE PROBLEM

PURPOSE

Two linked questions: how often did resolved tickets meet priority reference times, and how well can past arrivals predict daily help-desk demand?

TAKEAWAY Retrospective service analysis + 7- and 30-day volume forecasts.

2



UNDERSTAND THE DATA

DATA

66,691 source rows; 27,348 exact Ticket records. Scope: 2016-01-03 to 2023-03-14 (UTC). Earlier timestamps are excluded and flagged.

TAKEAWAY 2,628 calendar days; source identity verified, completeness unverified.

3



CLEAN, EXPLORE & ENGINEER

PROCESS

Build daily counts with past-only lags and calendar features. Compare training-only mutual-information selection and PCA for the Ridge challenger.

TAKEAWAY 344 empty dates are zero-filled under an unverified completeness assumption.

4



BUILD & COMPARE MODELS

METHOD

Compare seasonal naive, weekly ETS, XGBoost and MI/PCA Ridge using expanding-window backtests. Select on validation MAE; keep a later test year separate.

TAKEAWAY Selected model: ETS (weekly additive); 30-day validation MAE 4.39.

5



READ THE FORECAST RESULTS

RESULT

30-day horizon: test MAE 4.67 tickets/day; RMSE 7.20; WAPE 27.51%. This forecasts a daily path, not a single monthly total.

TAKEAWAY 90% interval coverage: 87.7% on historical tests; calibration is imperfect.

6



ASSESS SLA REFERENCE TIMES

SERVICE

3,333 of 16,735 eligible resolved tickets (19.92%) met the reference. Critical: 4h; High: 8h; Medium / Low: 24h, using elapsed UTC time.

TAKEAWAY Project-defined thresholds, not approved policy or contractual SLA compliance.

7



EXPLAIN, AUDIT & REFLECT

AUDIT

XGBoost SHAP: lag_28, lag_21, lag_14. Separate ETS state/residual diagnostics are in the report. No protected attributes: demographic fairness cannot be assessed.

TAKEAWAY Feature attribution shows model association, not causal demand drivers.

PACKAGE & DEMONSTRATE

- **Reproducible analysis**
Eight notebooks; shared outputs; report and two decks.
- **Model artifact + MLflow**
Saved model, metadata and tracked experiment results.
- **FastAPI + Power BI-ready exports**
Deployment proof of concept, not a live service.

BEFORE ANY OPERATIONAL USE

- **Resolve rights and coverage**
Retain CC BY 4.0; resolve mirror label and zero-day gaps.
- **Approve the service definitions**
Validate priority mapping, SLA clock and pause rules.
- **Re-test on current data**
Audit drift, interval coverage and operational segments.

RESEARCH PROTOTYPE | Historical results, not current-operations evidence.